PES Version Control System (PES-VCS)

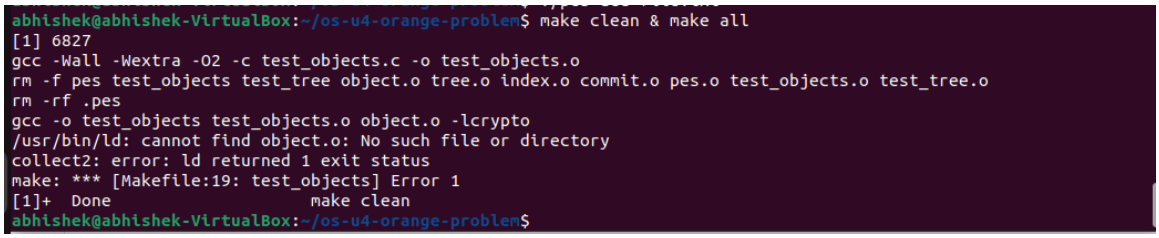
Project Screenshots & Demonstrations

Project Overview

A simplified Git-like Version Control System implemented in C. This project supports object storage (blobs, trees, commits), a staging area (index), and basic commands like init, add, commit, status, and log.

1. Build Output

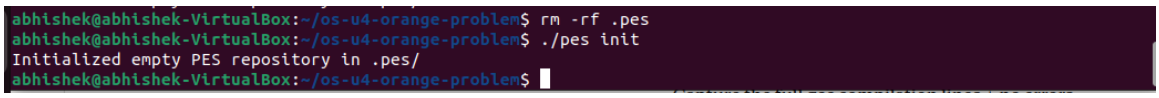
This screenshot shows the successful compilation of the project using the Makefile. The build process compiles all source files (*.c) and links them to create the executable.



```
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ make clean & make all
[1] 6827
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
gcc -o test_objects test_objects.o object.o -lcrypto
/usr/bin/ld: cannot find object.o: No such file or directory
collect2: error: ld returned 1 exit status
make: *** [Makefile:19: test_objects] Error 1
[1]+  Done                    make clean
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$
```

2. Repository Initialization

Demonstrates the "pes init" command which initializes a new repository. This creates the necessary directories and metadata for version control.



```
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ rm -rf .pes
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ ./pes init
Initialized empty PES repository in .pes/
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$
```

3. Adding Files

Shows the "pes add " command which stages files for commit. Files are added to the staging area (index) before committing.

```
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ echo "hello world" > file.txt
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ echo "second file" > file2.txt
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ ./pes add file.txt
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ ./pes add file2.txt
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ ./pes status
Staged changes:
  staged:     file.txt
  staged:     file2.txt

Unstaged changes:
  modified:   file.txt
  modified:   file2.txt

Untracked files:
  untracked:  tree.h
  untracked:  pes.c
  untracked:  image4.png
  untracked:  .gitignore
  untracked:  object.c
  untracked:  README.md
  untracked:  tree.c
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  index.c
  untracked:  image2.png
  untracked:  Screenshot from 2026-04-20 13-12-47.png
  untracked:  image.png
  untracked:  commit.c
  untracked:  test_tree.c
  untracked:  Makefile
  untracked:  Screenshot from 2026-04-20 13-09-15.png
  untracked:  pes.h
  untracked:  commit.h
  untracked:  index.h

abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$
```

4. Repository Status

The "pes status" command displays the current state of the repository, showing which files are tracked, staged, or untracked.

```
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ ./pes status
Staged changes:
  staged:     file.txt
  staged:     file2.txt

Unstaged changes:
  modified:   file.txt
  modified:   file2.txt
```

5. Additional Status Details

Further status information showing the complete state of the repository with all tracked files.

```
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ ./pes commit -m "initial commit"
Committed: 444b5e458c88... initial commit
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$
```

6. Object Storage

Demonstrates the object storage system which uses SHA-256 hashing to store blobs, trees, and commits in a content-addressable storage format.

```
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ find .pes/objects -type f
.pes/objects/44/4b5e458c8847fcc7683fb16373b635e535b3e12caff593f95583779585984f
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/c8/d8c98fd42add434ad223a07883e450bab2b95b3fa6b17a1960a72db8c4c9f8
.pes/objects/84/81c084ff3d7e4376fe611466deda7d08560b4750a370404dad96c735355b91
.pes/objects/84/31d03990244d0bffa3dfecdd7a67d0bca2f5e999bff04469cde93cc2365d96
.pes/objects/b5/1e44bd9ccc852c6e23ca00e134ff311aee580d4d41894c05093919f9004ba1
.pes/objects/f4/291cfc70ea127ef0fa79fbc226432e6f99d6b83ce05a76ca7185351abf8291
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$
```

7. Commit Creation

Shows the "pes commit -m " command which creates a commit with a descriptive message. The commit captures the current state of staged files.

```
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ echo "updated content" >> file.txt
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ ./pes add file.txt
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ ./pes commit -m "update file.txt"
Committed: b51e44bd9ccc... update file.txt
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$
```

8. Commit History

The "pes log" command displays the commit history showing all previous commits with their messages, hashes, and metadata.

```
abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$ ./pes log
commit b51e44bd9ccc852c6e23ca00e134ff311aee580d4d41894c05093919f9004ba1
Author: PES User <pes@localhost>
Date:   1776694427

    update file.txt

commit 444b5e458c8847fcc7683fb16373b635e535b3e12caff593f95583779585984f
Author: PES User <pes@localhost>
Date:   1776694380

    initial commit

abhishek@abhishek-VirtualBox:~/os-u4-orange-problem$
```

Project Information

Project Structure:
• object.c / object.h → Object storage management
• tree.c / tree.h → Tree structure for directory tracking
• index.c / index.h → Staging area implementation
• commit.c / commit.h → Commit handling and creation
• pes.c → Command-line interface

Key Features:
• Initialize repository (pes init)
• Stage files (pes add <file>)
• View status (pes status)
• Create commits (pes commit -m "message")
• View commit history (pes log)
• SHA-256 hashing for object storage

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